

YAR1 (P46683) — Transcription / MBF–SBF Membership Hypothesis: Focused Curation Report

Gene: YAR1 / *Saccharomyces cerevisiae* (NCBITaxon:559292) · UniProt **P46683** · 200 aa **Focus:** function_assignment · slug `transcriptional-regulation-and-mbf-sbf-membership` **Seed hypothesis:** Yar1 promotes RNA Pol II transcription or belongs to the MBF or SBF transcription complexes (to be adjudicated independently of DNA-binding transcription-activator activity and Rps3 chaperoning).

Executive Judgment

Verdict: REFUTED / OVER-ANNOTATED (for the transcription and MBF/SBF claims).

There is **no primary experimental evidence** that Yar1 promotes RNA Pol II transcription or is a subunit of the MBF (Mbp1–Swi6) or SBF (Swi4–Swi6) complexes. Every transcription-related GO annotation on P46683 carries evidence code **IBA** (Inferred from Biological Ancestor; GO_Central/PANTHER) — i.e., phylogenetically propagated, not observed. In contrast, all **experimentally supported** annotations (IDA/IMP/HDA) describe Yar1's role as a **dedicated chaperone of ribosomal protein Rps3 in 40S small-subunit biogenesis and its nuclear import**, backed by genetics, biochemistry, and a 2.8 Å co-crystal structure.

The source of the misannotation is traceable: since 1996 it has been noted that Yar1's ankyrin (ANK) repeats are most similar to those of the cell-cycle transcription factor **Swi6**. Yar1 and the SBF/MBF ankyrin proteins therefore fall into the same broad ankyrin-repeat PANTHER family (PTHR24198), and Swi6/SBF/MBF transcription terms propagate to Yar1 by IBA. **Crucially, in Swi6/Swi4/Mbp1 DNA binding is conferred by the APSES (KilA-N/winged-helix) domain, not by the ANK repeats** — and Yar1 possesses only ankyrin repeats, i.e., the generic protein–protein interaction module that does *not* bind DNA. Thus the seed's own Swi6/Cdc10 analogy, examined carefully, argues *against* the transcription claim rather than for it.

Most important caveat: absence of positive evidence is not proof of impossibility. A non-DNA-binding, sub-stoichiometric recruitment role cannot be formally excluded by the current data, but nothing in the primary literature or complex-composition datasets places Yar1 in MBF/SBF or at Pol II promoters. The parsimonious, evidence-based interpretation is a ribosome-biogenesis chaperone whose nuclear presence reflects Rps3 import, not transcription.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Co & lim
P 24021814 (Holzer, Ban, Klinge 2013)	Structural (co- crystal, direct)	Refutes transcription; supports chaperone	Primary molecular function of Yar1	2.8 Å Rps3– Yar1 structure; Yar1 keeps Rps3 soluble and mediates its nuclear import	<i>S. cerevisiae</i> , in vitro/ structural	Hi, str me de rea
P 15611164 (Loar et al 2004)	Mutant phenotype + interaction (direct)	Refutes transcription; supports 40S role	Whether Yar1 acts in transcription vs ribosome biogenesis	Yar1 binds Rps3 & Ltv1; Δyar1 has reduced 40S/ excess free 60S; RPS3 overexpression rescues	<i>S. cerevisiae</i> genetics/ biochem	Hi, Ya in pro
P 8675027 (Lycan et al 1996)	Sequence/ original characterization	Qualifies (explains the artifact)	Origin of Swi6/SBF link	Yar1 ANK repeats "most similar to... Swi6"; non- essential; slow- growth/cold- sensitive Δyar1; no essential TF phenotype	<i>S. cerevisiae</i>	Hi, sin cla au no de TF
P 16888326 (Seiser et al 2006)	Genetics/ pathway (indirect)	Supports 40S/ export context	Pathway context of Yar1 partners	Ltv1 (Yar1 partner) required for 40S nuclear export via Crm1	<i>S. cerevisiae</i>	Me Co ab no dir
		Refutes direct	Evidence basis of TF/	MBF complex, SBF complex, DNA-binding	GO_Central/ PANTHER	Hi, ph

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Co & lim
UniProt P46683 GO set (database)	Database/ annotation provenance	transcription support	MBF/SBF terms	TF activator activity, +reg Pol II transcription are all IBA ; all experimental terms are ribosome/ localization		inf no
InterPro/ PANTHER (P46683) (computational)	Structural/ evolutionary	Refutes DNA-binding capacity	Does Yar1 have a DNA- binding domain?	Only ANK signatures (IPR002110, Pfam PF12796, PANTHER PTHR24198); no APSES/ KilA-N/ winged-helix domain	Public bioinformatics	Hi an bin
EBI Complex Portal CPX-946, CPX-950 (database, curated)	Interaction/ complex composition	Refutes MBF/SBF membership	Is Yar1 a subunit of SBF or MBF?	SBF (CPX-946) = Swi4 + Swi6; MBF (CPX-950) = Mbp1 + Swi6; Yar1/P46683 absent from both	<i>S. cerevisiae</i> , manually curated	Hi Au co de
UniProt domain comparison (computational)	Structural/ evolutionary	Refutes/ qualifies	Does Yar1 share the DNA-binding module of SBF/MBF?	Swi4/Mbp1 carry APSES- type HTH + H- T-H DNA- binding motif; Yar1 (200 aa) has only 2 ANK repeats — the	Public bioinformatics	Hi lac DN co sul are

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Co & lim
				generic interaction module		
IntAct interactome (P46683) (database, interaction)	Interaction	Refutes transcription/ complex link	Does Yar1 physically interact with SBF/MBF subunits?	15 records; partners = Rps3, Ltv1 + Hsp70/Hsp40 chaperones (Ssa1/2, Ssb1, Sse1, Sis1, Ydj1, Hsp78/26, Erj5, Ess1, Sfm1); no Swi4/Swi6/ Mbp1	<i>S. cerevisiae</i> , curated PPIs	Hi int ev a tra rol

GO Curation Implications (leads requiring curator verification)

GO term	Aspect	Current evidence	Recommended action (lead)
GO:0001228 DNA-binding transcription activator activity, Pol II-specific	MF	IBA only; no APSES domain	Remove / do not propagate. Not supported; contradicted by domain architecture
GO:0030907 MBF transcription complex	CC	IBA only; no complex-composition data	Remove / mark NOT. No primary evidence Yar1 is an MBF subunit
GO:0033309 SBF transcription complex	CC	IBA only	Remove / mark NOT. No primary evidence Yar1 is an SBF subunit
GO:0045944 positive regulation of transcription by Pol II	BP	IBA only	Remove / treat as non-core. No direct evidence; any effect is downstream of ribosome biogenesis
GO:0042274 ribosomal small subunit biogenesis	BP	IMP (experimental)	Retain (core).
GO:0000056 ribosomal small subunit export from nucleus	BP	IMP	Retain.
GO:0005634 nucleus / GO:0005737 cytoplasm	CC	IDA/HDA	Retain. Nuclear signal reflects Rps3 import, not TF role
(candidate addition) protein chaperone / unfolded-protein binding on Rps3	MF	Direct structure 	Consider adding a chaperone MF term (e.g., unfolded/nascent-protein binding) — more informative than "protein binding"

Bottom line: The four IBA transcription terms are over-annotations and are the strongest candidates for removal or a NOT qualifier; the ribosome-biogenesis terms are the core, experimentally supported function.

Mechanistic Scope

- **Direct molecular activity:** Yar1 is a small (200 aa) two-ANK-repeat protein that acts as a **dedicated chaperone binding ribosomal protein Rps3**, holding it in a soluble, import-competent state (co-crystal structure,

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- **Cellular process:** couples Rps3 folding/solubility to **nuclear import and incorporation into the assembling 40S subunit**

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- **Downstream/indirect only:** stress sensitivity, growth/cold-sensitivity of Δ yar1, and any perturbation of gene-expression programs are **secondary consequences of impaired 40S biogenesis**, not evidence of a direct transcriptional function. Nuclear localization is expected for an Rps3 import chaperone and does not imply promoter association.

Complex Composition & Domain Comparison (computed, Iteration 2)

EBI Complex Portal (manually curated):

Complex	Accession	Subunits	Yar1 present?
SBF transcription complex	CPX-946	Swi4 (P25302) + Swi6 (P09959)	No
MBF ("MBP") transcription complex	CPX-950	Mbp1 (P39678) + Swi6 (P09959)	No

UniProt domain architecture:

Protein	Length	APSES/HTH DNA-binding domain	ANK repeats	Role
Yar1 (P46683)	200 aa	None	2	Rps3 chaperone (not a TF)
Swi6 (P09959)	803 aa	None	5	Shared regulatory subunit (SBF+MBF)
Swi4 (P25302)	1093 aa	APSES-HTH (37–147) + H-T-H (71–92)	2	SBF DNA-binding subunit
Mbp1 (P39678)	833 aa	APSES-HTH (5–111) + H-T-H (36–57)	2	MBF DNA-binding subunit

Interpretation: Yar1 is absent from both curated complexes. The seed's Swi6/Cdc10 precedent (a subunit lacking APSES) is technically real — Swi6 indeed has no APSES domain — but it does **not** transfer to Yar1: Swi6 is an 803-aa subunit with direct experimental complex-membership evidence, whereas Yar1 is a 200-aa protein with none. The **only** feature Yar1 shares with any SBF/MBF subunit is the generic ANK repeat, and Yar1's ANK repeats are structurally documented to bind **Rps3**

([P 24021814](#)),

not Swi4/Swi6/Mbp1. Shared ANK possession — the driver of the IBA propagation — is not evidence of shared complex membership.

Conflicts and Alternatives

- **Paralog/family confusion (primary artifact):** Yar1 shares the ankyrin-repeat module with genuine SBF/MBF subunit **Swi6**. Broad PANTHER family PTHR24198 grouping drives IBA propagation of Swi6-derived transcription terms.
- **Domain-logic conflict:** DNA binding in APSES transcription factors (Swi4/Mbp1) resides in the **APSES/KiLA-N domain**, absent from Yar1; Swi6 itself contributes regulation via ANK-mediated protein interactions but is not the DNA-contacting subunit. So the "ANK repeats ≈ transcription factor" inference is unsound.
- **Alternative interpretation fully consistent with all data:** Yar1 = ribosome-assembly chaperone; no competing primary dataset supports an MBF/SBF or Pol II role.

- The seed's Swi6/Cdc10 precedent (non-DNA-binding complex membership) is a valid *general* caution but has **no gene-specific support** for Yar1.

Knowledge Gaps

1. **Direct complex-composition/interaction test.** *Checked:* Yar1 absent from curated MBF/SBF (Complex Portal); Yar1's full IntAct interactome (Rps3, Ltv1, Hsp70/Hsp40 chaperones) contains **no** Swi4/Swi6/Mbp1. *Matters:* IBA terms would need a real subunit assignment to stand; none exists. *Status:* **largely resolved against the hypothesis.** *Further confirmation:* Yar1-TAP MS would be expected to recover Rps3/Ltv1/chaperones, not SBF/MBF.
 2. **Promoter occupancy.** *Checked:* none found. *Matters:* required for "positive regulation of Pol II transcription." *Resolve:* Yar1 ChIP-seq/ChIP-exo — expect no specific promoter binding.
 3. **Transcriptome specificity.** *Checked:* Δ yar1 phenotypes are ribosome-linked. *Matters:* distinguishes direct TF role from indirect ribosome-biogenesis effects. *Resolve:* nascent-transcription (4tU/GRO-seq) in Δ yar1 vs 40S-biogenesis controls — expect shared, indirect signature.
 4. **Precise provenance of each IBA call.** *Checked:* confirmed all four are IBA/GO_Central. *Matters:* pinpoints the ancestral node for a targeted GO_Central fix. *Resolve:* inspect the PTHR24198 reference tree annotation node.
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Discriminating Tests

- **Yar1-TAP affinity-MS** in exponentially growing cells: predicted interactors Rps3, Ltv1, import factors — **not** Swi4/Swi6/Mbp1.
- **Reciprocal SBF/MBF pulldowns** (Swi6-, Swi4-, Mbp1-TAP): predicted absence of Yar1.
- **Yar1 ChIP-seq** at cell-cycle (SCB/MCB) promoters: predicted no enrichment.
- **Domain rescue logic:** Δ yar1 growth defect is rescued by *RPS3* overexpression (already shown,

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— a transcription-factor loss would not be rescued this way.

- **Sequence/structure:** HHpred/InterProScan for an APSES domain in Yar1 — predicted none (consistent with our InterPro result).

Curation Leads (verify before applying)

- **Action:** Downgrade/remove the four IBA transcription annotations (GO:0001228, GO:0030907, GO:0033309, GO:0045944) as phylogenetic over-annotation from a shared ankyrin-repeat family; consider NOT qualifiers if IBA cannot be locally suppressed.
- **Root-cause fix:** Flag PANTHER **PTHR24198** reference-tree node for GO_Central review — Swi6-derived transcription terms should not descend to the Yar1 chaperone lineage.
- **Retain as core:** GO:0042274 (small-subunit biogenesis), GO:0000056 (small-subunit nuclear export), nucleus/cytoplasm CC.
- **Candidate addition (MF):** a chaperone/unfolded-protein-binding term for Rps3 (avoid generic "protein binding"), supported by

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- **Candidate references to cite in the review:** PMID **24021814** (co-crystal, chaperone mechanism), PMID **15611164** (40S genetics/interactions), PMID **8675027** (origin of Swi6 similarity; explicitly note no TF activity was shown).
- **Suggested reviewer question:** "Is there any dataset placing Yar1 in MBF/SBF or at Pol II promoters independent of its ankyrin-repeat homology to Swi6?" (Expected: no.)

Provenance

Computed evidence (executed code + outputs retained in the job log): - **Iter 1:** (1) UniProt P46683 REST pull listing GO evidence codes — all transcription terms = IBA, all experimental terms = ribosome/localization; (2) InterPro/PANTHER signature pull for P46683 — ankyrin-repeat signatures only (IPR002110, PF12796, PTHR24198), no APSES/DNA-binding domain. - **Iter 2:** (3) EBI Complex Portal search + composition of CPX-946 (SBF = Swi4+Swi6) and CPX-950 (MBF = Mbp1+Swi6) — Yar1/P46683 absent from both; (4) UniProt domain comparison Yar1 vs Swi6/Swi4/Mbp1 — Yar1 has only ANK repeats, Swi4/Mbp1 carry APSES-HTH + H-T-H DNA-

binding motifs. - **Iter 3:** (5) IntAct/PSICQUIC interactome of P46683 (15 records) — partners = Rps3, Ltv1 + Hsp70/Hsp40 chaperone network (Ssa1/2, Ssb1, Sse1, Sis1, Ydj1, Hsp78/26, Erj5, Ess1, Sfm1); no Swi4/Swi6/Mbp1.

Literature via PubMed (PMIDs 8675027, 15611164, 16888326, 24021814).

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